

Request for Project Extension



◆ 1. Create Database

```
SQL  
create database student;
```

- **create** → command to make something new
- **database** → collection of tables
- **student** → name of the database

👉 Creates a database named **student**

◆ 2. Create Table: academic

```
SQL  
create table student.academic(
```

- **create table** → create a new table

- `student.academic` → table `academic` inside `student` database

```
SQL
student_id int,
```

- `student_id` → column name
- `int` → integer type (numbers only)

```
SQL
firstname varchar(255),
```

- `firstname` → column name
- `varchar(255)` → text up to 255 characters

```
SQL
remarks varchar(255) default 'Can do better',
```

- `remarks` → column name
- `default 'Can do better'` → if no value is given, this text is inserted automatically

```
SQL
marks int,
```

- `marks` → column name
- `int` → numeric value

```
SQL
check (marks>35)
```

- `check` → constraint (rule)
- `(marks > 35)` → marks must be greater than 35

👉 If marks $\leq$ 35 → ❌ insert fails

```
SQL  
) ;
```

👉 Ends table creation

◆ 3. Create Table: sports

```
SQL  
create table student.sports(
```

👉 Create another table

```
SQL  
sport_id int,  
sport_name varchar(255),  
student_id int,
```

- **sport_id** → sport number
- **sport_name** → sport name
- **student_id** → reference to student

```
SQL  
constraint abc check(sport_id != 0)
```

- **constraint abc** → naming the constraint as **abc**
- **check(sport_id != 0)** → sport_id cannot be 0

◆ 4. Insert Data (valid)

```
SQL  
insert into student.academic (student_id,firstname,marks)  
values (2, 'B', 40);
```

- **insert into** → add data
- **(student_id, firstname, marks)** → columns

- **values (...)** → actual data

👉 remarks will be default → "Can do better"

◆ 5. Alter Table (Set Default)

```
SQL
alter table student.academic
alter marks set default 36;
```

- **alter table** → modify table
- **alter marks** → change column
- **set default 36** → default marks = 36

◆ 6. Insert (using default)

```
SQL
insert into student.academic (student_id,firstname)
values (3, 'C');
```

👉 marks will automatically be **36**

◆ 7. Remove Default

```
SQL
alter table student.academic
alter marks drop default;
```

- removes default value from marks

◆ 8. Insert (Invalid - fails)

```
SQL
insert into student.academic
values (2, 'B', 'GOOD', 33);
```

👉 ❌ Fails because:

- **marks = 33**

- check condition = `marks > 35`

◆ 9. Insert (Valid)

```
SQL
insert into student.academic
values (2, 'B', 'GOOD', 36);
```

👉 ✅ Works (36 > 35)

◆ 10. Insert into sports (Invalid)

```
SQL
insert into student.sports
values (0, 'cricket', 10);
```

👉 ❌ Fails

- because `sport_id != 0` condition

◆ 11. Insert into sports (Valid)

```
SQL
insert into student.sports
values (1, 'cricket', 10);
```

👉 ✅ Works

◆ 12. Add New Check Constraint

```
SQL
alter table student.academic
add check (student_id < 100);
```

👉 Now:

- student_id must be **less than 100**

◆ 13. Insert (Invalid)

```
SQL
insert into student.academic
values (101, 'c', 'GOOD', 60);
```

👉 ❌ Fails

- because student_id = 101 (>100)

◆ 14. Insert (Invalid)

```
SQL
insert into student.academic
values (101, 'c', 'GOOD', 6);
```

👉 ❌ Fails for 2 reasons:

- student_id > 100
- marks < 35

◆ 15. Drop Constraint

```
SQL
alter table student.sports
drop check abc;
```

- removes constraint named abc

👉 Now sport_id can be 0

✅ Final Understanding (Very Important)

Constraints used:

- default → auto value
- check → condition rule

- **constraint name** → naming rules

Pro Tip (Interview)

- **CHECK** ensures **data validation**
- **DEFAULT** ensures **automatic value**
- **ALTER** modifies structure

If you want, I can:

- ✓ Convert this into **notes (PDF style)**
- ✓ Give **interview questions from this**
- ✓ Give **practice questions like this**

You said:

0 65 21:49:39 insert into student.academic values (2,'B','GOOD',33) Error Code: 3819. Check constraint 'academic_chk_1' is violated. 0.000 sec

ChatGPT said:

This error is actually **correct behavior** 👍 — your database is protecting invalid data.

Error Message

```
Error Code: 3819. Check constraint 'academic_chk_1' is violated.
```

What it means:

- **check constraint** = rule you defined earlier
- **'academic_chk_1'** = auto-generated name of the check
- **violated** = rule is broken ❌

Why this error happened

You created this rule:

```
SQL
check (marks > 35)
```

Now you tried:

```
SQL
insert into student.academic
values (2, 'B', 'GOOD', 33);
```

👉 Problem:

- `marks = 33`
- Rule = `marks > 35`

So:

```
33 > 35 ❌ FALSE
```

👉 That's why MySQL blocked the insert.

✅ How to Fix It

✔ Option 1: Insert valid data

```
SQL
insert into student.academic
values (2, 'B', 'GOOD', 36);
```

✔ Option 2: Change the condition (if needed)

If you want to allow marks like 33:

```
SQL
alter table student.academic
drop check academic_chk_1;
```

Then add new rule:


```
SQL
alter table student.academic
add check (marks >= 33);
```

✔ Option 3: Remove constraint completely

```
SQL
alter table student.academic
drop check academic_chk_1;
```

👉 Now any marks can be inserted

⚡ Important Concept

- **CHECK** constraint = **data validation rule**
- Database **prevents wrong data automatically**
- This is why SQL is powerful 🙌

💡 Quick Summary

Marks	Result
40	✔ Allowed
36	✔ Allowed
35	✗ Not allowed
33	✗ Not allowed

If you want, I can give:

- 🔥 Real interview questions on constraints
- 🔥 More tricky SQL errors like this (very useful for exams & jobs)

You said:

0 71 21:51:16 insert into student.academic values (101,'c','GOOD',6) Error Code: 3819. Check constraint 'academic_chk_1' is violated. 0.000 sec

ChatGPT said:

Good 👍 this is another **constraint error**, let's break it clearly.

🔴 Your Query

```
SQL
insert into student.academic
values (101, 'c', 'GOOD', 6);
```

❌ Why Error Occurred

You have **TWO check constraints** on this table:

1 Marks Constraint

```
SQL
check (marks > 35)
```

👉 Your value:

```
SQL
marks = 6 ❌ (fails)
```

2 Student ID Constraint

```
SQL
check (student_id < 100)
```

👉 Your value:

```
SQL
student_id = 101 ❌ (fails)
```

🌟 Final Result

👉 BOTH conditions failed, but MySQL shows only one error:

```
SQL
Error Code: 3819. Check constraint 'academic_chk_1' is
violated.
```

👉 This usually refers to the **first check constraint (marks > 35)**

✅ How to Fix

✔ Correct Data (Follow both rules)

```
SQL
insert into student.academic
values (99, 'c', 'GOOD', 60);
```

✔ student_id < 100

✔ marks > 35

⚡ Important Concept

👉 A row is inserted **ONLY IF ALL constraints are satisfied**

Think like this:

```
Condition 1 AND Condition 2 must be TRUE
```

If even **one is false** → ❌ **INSERT fails**

💡 Quick Table

student_id	marks	Result
101	60	❌ (id fails)
50	6	❌ (marks fails)
101	6	❌ (both fail)
99	60	✅ success

🔥 Pro Tip (Interview)

👉 SQL constraints work like **filters**

👉 Data must pass **ALL filters**

If you want, next I can teach:

- ✓ Primary Key vs Unique vs Check (very important)
- ✓ Real interview SQL questions based on this table
- ✓ Tricky constraint questions (asked in exams)

You said:

```
create database student; create table student.academic( student_id int,
firstname varchar(255), remarks varchar(255) default 'Can do better', marks int,
check (marks>35) ); create table student.sports( sport_id int, sport_name
varchar(255), student_id int, constraint abc check(sport_id != 0) ); insert into
student.academic (student_id,firstname,marks) values (2,'B',40); alter table
student.academic alter marks set default 36; insert into student.academic
(student_id,firstname) values (3,'C'); alter table student.academic alter marks
drop default; insert into student.academic values (2,'B','GOOD',33); insert into
student.academic values (2,'B','GOOD',36); insert into student.sports values
(0,'cricket',10); insert into student.sports values (1,'cricket',10); alter table
student.academic add check (student_id<100); insert into student.academic
values (101,'c','GOOD',60); insert into student.academic values
(101,'c','GOOD',6); alter table student.sports drop check abc; explain in
comment each word and line
```

ChatGPT said:

Here is your **same SQL code with detailed comments (line-by-line + word meaning)** 👉

```
SQL
-- Create a new database named 'student'
create database student;

-- Create a table named 'academic' inside 'student' database
create table student.academic(

    -- Column: student_id (integer type)
    student_id int,

    -- Column: firstname (string up to 255 characters)
    firstname varchar(255),

    -- Column: remarks (string) with default value
    -- If no value is given, 'Can do better' will be inserted
    automatically
    remarks varchar(255) default 'Can do better',

    -- Column: marks (integer type)
    marks int,

    -- CHECK constraint: marks must be greater than 35
    -- If marks <= 35 → insertion will fail
    check (marks > 35)
);

-- Create another table 'sports'
create table student.sports(

    -- Column: sport_id (integer)
    sport_id int,

    -- Column: sport_name (string)
    sport_name varchar(255),

    -- Column: student_id (integer, used to relate student)
    student_id int,

    -- Named constraint 'abc'
    -- CHECK condition: sport_id must NOT be 0
    constraint abc check (sport_id != 0)
);

-- Insert data into academic table
-- Only student_id, firstname, marks are provided
-- remarks will take default value 'Can do better'
insert into student.academic (student_id, firstname, marks)
values (2, 'B', 40);

-- Modify table: change 'marks' column default value to 36
alter table student.academic
alter marks set default 36;
```

```
-- Insert without marks → default value (36) will be used
insert into student.academic (student_id, firstname)
values (3, 'C');

-- Remove default value from 'marks' column
alter table student.academic
alter marks drop default;

-- Try inserting full row
-- marks = 33 → violates CHECK (marks > 35)
-- ❌ This will give ERROR
insert into student.academic
values (2, 'B', 'GOOD', 33);

-- Valid insert
-- marks = 36 → satisfies CHECK condition
-- ✅ This will work
insert into student.academic
values (2, 'B', 'GOOD', 36);

-- Insert into sports
-- sport_id = 0 → violates constraint abc (sport_id != 0)
-- ❌ ERROR
insert into student.sports
values (0, 'cricket', 10);

-- Valid insert into sports
-- sport_id = 1 → satisfies condition
-- ✅ Works
insert into student.sports
values (1, 'cricket', 10);

-- Add a new CHECK constraint on academic table
-- student_id must be less than 100
alter table student.academic
add check (student_id < 100);

-- Insert with student_id = 101 → violates new constraint
-- ❌ ERROR
insert into student.academic
values (101, 'c', 'GOOD', 60);

-- Insert with two violations:
-- student_id = 101 (>100 ❌)
-- marks = 6 (<35 ❌)
-- ❌ ERROR
insert into student.academic
values (101, 'c', 'GOOD', 6);

-- Drop (remove) constraint named 'abc' from sports table
-- After this, sport_id can be 0
alter table student.sports
drop check abc;
```

Final Understanding (Simple)

- **create** → create something
- **database** → collection of tables
- **table** → structure to store data
- **insert** → add data
- **alter** → modify table
- **default** → auto value
- **check** → condition (rule)
- **constraint** → named rule

Key Concept (VERY IMPORTANT)

 SQL will **NOT** allow wrong data

All conditions must be TRUE:

```
marks > 35 AND student_id < 100
```

If any one fails →  ERROR

If you want next:

- ✓ I can give **MCQs + interview questions from this**
- ✓ Or create **practice questions like exam level**

You said:

```
create database student; create table student.academic( student_id int,
firstname varchar(255), remarks varchar(255) default 'Can do better', marks int,
check (marks>35) ); create table student.sports( sport_id int, sport_name
varchar(255), student_id int, constraint abc check(sport_id != 0) ); insert into
student.academic (student_id,firstname,marks) values (2,'B',40); alter table
student.academic alter marks set default 36; insert into student.academic
(student_id,firstname) values (3,'C'); alter table student.academic alter marks
drop default; insert into student.academic values (2,'B','GOOD',33); insert into
student.academic values (2,'B','GOOD',36); insert into student.sports values
```

(0,'cricket',10); insert into student.sports values (1,'cricket',10); alter table student.academic add check (student_id<100); insert into student.academic values (101,'c','GOOD',60); insert into student.academic values (101,'c','GOOD',6); alter table student.sports drop check abc; explain in comment each word and line explain easy way

ChatGPT said:

Got it 👍 — I'll explain your code in a **very easy + simple way (like beginner level)** with comments.

```
SQL
-- Create a database named student
create database student;
```

👉 Makes a new database (storage area)

```
SQL
-- Create table academic inside student database
create table student.academic(
```

👉 Create table to store student data

```
SQL
    student_id int,
```

👉 student id (only numbers)

```
SQL
    firstname varchar(255),
```

👉 student name (text)

```
SQL
    remarks varchar(255) default 'Can do better',
```

👉 remarks column

👉 if you don't give value → it will automatically store "**Can do better**"


```
SQL
    marks int,
```

👉 marks column (number)

```
SQL
    check (marks > 35)
```

👉 RULE: marks must be greater than 35

👉 if marks = 35 or less ❌ error

```
SQL
);
```

👉 table created

```
SQL
-- Create sports table
create table student.sports(
```

👉 another table for sports

```
SQL
    sport_id int,
```

👉 sport id number

```
SQL
    sport_name varchar(255),
```

👉 sport name (like cricket)

```
SQL
    student_id int,
```

👉 which student plays sport

```
SQL
constraint abc check (sport_id != 0)
```

- 👉 RULE: sport_id cannot be 0
- 👉 abc is just the name of this rule

```
SQL
);
```

```
SQL
-- Insert data (valid)
insert into student.academic (student_id, firstname, marks)
values (2, 'B', 40);
```

- 👉 works ✅
- 👉 marks > 35 ✔️
- 👉 remarks auto = "Can do better"

```
SQL
-- Set default marks = 36
alter table student.academic
alter marks set default 36;
```

- 👉 if marks not given → 36 will be stored

```
SQL
-- Insert without marks
insert into student.academic (student_id, firstname)
values (3, 'C');
```

- 👉 marks = 36 (default) ✅

```
SQL
-- Remove default value
alter table student.academic
alter marks drop default;
```

- 👉 now marks must be given manually

```
SQL
-- Insert wrong data
insert into student.academic
values (2, 'B', 'GOOD', 33);
```

👉 ❌ ERROR

👉 because marks = 33 (not > 35)

```
SQL
-- Insert correct data
insert into student.academic
values (2, 'B', 'GOOD', 36);
```

👉 ✅ works

```
SQL
-- Insert wrong sport_id
insert into student.sports
values (0, 'cricket', 10);
```

👉 ❌ ERROR

👉 because sport_id cannot be 0

```
SQL
-- Correct insert
insert into student.sports
values (1, 'cricket', 10);
```

👉 ✅ works

```
SQL
-- Add new rule
alter table student.academic
add check (student_id < 100);
```

👉 now student_id must be less than 100

```
SQL
-- Insert wrong student_id
insert into student.academic
values (101, 'c', 'GOOD', 60);
```

👉 ❌ ERROR

👉 because 101 > 100

```
SQL
-- Insert totally wrong data
insert into student.academic
values (101, 'c', 'GOOD', 6);
```

👉 ❌ ERROR (2 reasons)

- student_id > 100
- marks < 35

```
SQL
-- Remove constraint abc
alter table student.sports
drop check abc;
```

👉 now sport_id can be 0 also

🧠 Super Simple Summary

◆ Rules you created:

- marks must be > 35
- sport_id must not be 0
- student_id must be < 100

◆ Important Logic:

👉 SQL checks like this:

```
ALL CONDITIONS must be TRUE
```

If even one fails:

```
❌ INSERT FAILED
```

Real-Life Example

Think like:

- marks rule = pass exam
- id rule = valid student

If any rule breaks →  entry rejected

If you want next:

- ✓ I can give **short notes for revision**
- ✓ OR **interview questions from this topic** 

You said:

create database student; create table student.academic(student_id int, firstname varchar(255), remarks varchar(255) default 'Can do better', marks int, check (marks>35)); create table student.sports(sport_id int, sport_name varchar(255), student_id int, constraint abc check(sport_id != 0)); insert into student.academic (student_id,firstname,marks) values (2,'B',40); alter table student.academic alter marks set default 36; insert into student.academic (student_id,firstname) values (3,'C'); alter table student.academic alter marks drop default; insert into student.academic values (2,'B','GOOD',33); insert into student.academic values (2,'B','GOOD',36); insert into student.sports values (0,'cricket',10); insert into student.sports values (1,'cricket',10); alter table student.academic add check (student_id<100); insert into student.academic values (101,'c','GOOD',60); insert into student.academic values (101,'c','GOOD',6); alter table student.sports drop check abc; explain in comment each word and line easy way to understand each word expalin in comment

ChatGPT said:

Here is your **same SQL code with very simple word-by-word comments** 
(easy to understand)

```
SQL
-- create = make new
-- database = storage area
-- student = database name
create database student;
```

```
SQL
-- create = make new
-- table = structure to store data
-- student.academic = academic table inside student database
create table student.academic(

    -- student_id = column name
    -- int = integer (number)
    student_id int,

    -- firstname = column name
    -- varchar(255) = text up to 255 characters
    firstname varchar(255),

    -- remarks = column name
    -- default = automatic value if not given
    -- 'Can do better' = default text
    remarks varchar(255) default 'Can do better',

    -- marks = column name
    -- int = number
    marks int,

    -- check = rule/condition
    -- marks > 35 = marks must be greater than 35
    check (marks > 35)

);
```

```
SQL
-- create another table
create table student.sports(

    -- sport_id = number column
    sport_id int,

    -- sport_name = text column
    sport_name varchar(255),

    -- student_id = number column
    student_id int,

    -- constraint = giving name to rule
    -- abc = name of rule
    -- check = condition
    -- sport_id != 0 = sport_id should not be 0
    constraint abc check (sport_id != 0)

);
```

SQL

```
-- insert = add data
-- into = inside table
-- student.academic = table name
-- (columns) = where data will go
insert into student.academic (student_id, firstname, marks)

-- values = actual data
values (2, 'B', 40);
-- remarks not given → default used
```

SQL

```
-- alter = change table
-- set default = set automatic value
alter table student.academic
alter marks set default 36;
```

SQL

```
-- insert without marks
insert into student.academic (student_id, firstname)
values (3, 'C');
-- marks = 36 (default)
```

SQL

```
-- drop default = remove automatic value
alter table student.academic
alter marks drop default;
```

SQL

```
-- insert full row
insert into student.academic
values (2, 'B', 'GOOD', 33);

-- ERROR ❌
-- because marks = 33
-- rule = marks > 35
```

```
SQL
-- correct insert
insert into student.academic
values (2, 'B', 'GOOD', 36);

-- works ✓
```

```
SQL
-- insert into sports
insert into student.sports
values (0, 'cricket', 10);

-- ERROR ✗
-- because sport_id = 0
-- rule = sport_id != 0
```

```
SQL
-- correct insert
insert into student.sports
values (1, 'cricket', 10);

-- works ✓
```

```
SQL
-- add new check rule
alter table student.academic
add check (student_id < 100);

-- now student_id must be < 100
```

```
SQL
-- insert wrong student_id
insert into student.academic
values (101, 'c', 'GOOD', 60);

-- ERROR ✗
-- student_id > 100
```



```
SQL
-- insert totally wrong data
insert into student.academic
values (101, 'c', 'GOOD', 6);

-- ERROR ❌
-- reason 1: student_id > 100
-- reason 2: marks < 35
```

```
SQL
-- drop = remove
-- check abc = remove constraint named abc
alter table student.sports
drop check abc;

-- now sport_id can be 0
```



Super Easy Summary

Keywords meaning:

- **create** → make
- **database** → storage
- **table** → structure
- **insert** → add data
- **alter** → change
- **default** → automatic value
- **check** → rule
- **constraint** → named rule
- **drop** → remove



Golden Rule

```
SQL
-- ALL rules must be true
```

👉 If any condition fails:

```
SQL
```

```
-- INSERT will fail ❌
```

If you want next:

- ✅ I can make **short notes for exam revision (1 page)**
- ✅ OR give **important interview questions from this topic** 🚀

ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#).